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Inventor(s)

Godwin, JR.; James Patrick

Automatic Tarp Control System and Method for Dump Trucks

Abstract

A tarp control system for a dump truck controls a motorized spool to automatically extend and retract the tarp in response to a non-continuous input from a user. A rotary encoder attached to the motorized spool outputs the spool's motion to a controller. In response to user input, the controller controls the motor of the motorized spool to rotate in the proper direction to extend or retract the tarp. Rearward-biased tarp arms pull the tarp over the dump truck bed as the motorized spool unwinds. When the tarp reaches a (previously calibrated) fully extended position, the controller stops motion of the motorized spool. An inverse process retracts the tarp automatically in response to a non-continuous user input. The tarp control system allows a dump truck operator to initiate, then ignore, the tarp operation and concentrate on driving or other tasks, as the tarp extends or retracts automatically.

Inventors: Godwin, JR.; James Patrick (Dunn, NC)

Applicant: Intellectual Property Group, LLC (Dunn, NC)

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Background/Summary

FIELD OF INVENTION

[0001] The present invention relates generally to dump trucks, and in particular to a system and method of automatically controlling the extension and retraction of a tarp over the dump truck bed.

BACKGROUND

[0002] The global market for dump trucks in 2022 was \$8.1B, and is projected to increase to \$15B by 2030. Dump trucks are used in mining, construction, waste management, snow and ice removal, and other industries. Growth in the dump truck market is driven by increased exploration in the mining industry, and renewed (post-COVID) investment in transportation infrastructure and industrial, commercial, and residential building construction. Investment in new dump trucks is required to meet stringent emissions and other environmental controls, and to take advantage of technological advances.

[0003] FIG. 1 depicts a typical dump truck **10**. The dump truck **10** comprises a cab **12** and articulated bed **14**. The bed **14** of a typical dump truck **10** comprises four sides of a six-sided rectangular shape: a front wall, two side walls, and a floor. The top of the bed **14** is open, so the bed **14** may be filled from a tippie, front loader, or the like. The rear of the bed **14** is also open. A tail gate **16**, which is hinged at the top and releasably connected to the side walls near the bottom, covers the rear opening. The bed **14** is pivotally connected to the dump truck frame at or near its rear, and one or more hydraulic pistons connect the bed **14** to the truck frame at or near the front of the bed **14**. As the hydraulic piston(s) are extended, the bed **14** raises. If the tail gate **16** is released from the side walls, it remains vertical as the bed **14** raises, hanging from its pivotal connection at the top, and the load (sand, gravel, salt, etc.) slides out of the bed **14** and under the tail gate **16**, or is “dumped,” due to gravity.

[0004] In many jurisdictions, a cloth, plastic, or similar cover—referred to herein as a “tarp” **18**—is required to cover the top of the bed **14**, preventing any of the load from escaping out of the open bed top, such as by wind or jostling due to potholes and the like, during transit. Such loss not only reduces the delivered load, but may present a hazard to road traffic.

[0005] The tarp **18** is wound on a motorized spool **20** at the top of the front wall of the dump truck bed **14**, as shown in FIG. 1. Tarp arms **22** are pivotally mounted at their bottom ends to the dump truck bed **14**, at or near the midpoint of the bed **14** length. The tarp arms **22** are biased towards an extended position, towards the rear of the dump truck bed **14**. The tarp **18** is extended and retracted by controlling the motorized spool **20**. Typically, the two positions of a rocker switch actuate the motorized spool **20** in different directions, to either extend or retract the tarp **18**. FIG. 2 shows the tarp **18** in an intermediary position, as the motorized spool **20** is turning to unwind the tarp **18**, and the tarp arms **22** draw the leading edge of the tarp **18** towards the rear of the bed **14**. When the tarp **18** reaches the extent of its deployment or retraction, the operator should release the rocker switch to stop further movement of the motorized spool **20**. FIG. 3 shows the tarp **18** fully extended.

Operators typically monitor movement of the tarp arms **22** using the truck's external mirrors. When extended, the rearward bias of the tarp arms **22** places the tarp **18** in tension, which prevents it from “flapping” in the wind during transit.

[0006] Although this simple system works well under controlled circumstances, it may present a significant safety hazard in practice. As one example, at a quarry or mine, the tarp **18** must be retracted to expose the dump truck bed **14** so it may be filled with ore, aggregate, or other load. Regulations may require that the tarp **18** be extended prior to the dump truck **10** exiting the site and entering public roads. Because sitting still at the site exit to extend the tarp **18** is “dead time,” dump truck operators may extend the tarp **18** during transit from the location of the dump truck **10** loading to the site exit, so the tarp **18** is extended by the time they reach the exit, and they can proceed to transport the load. This requires that the operator hold down the rocker switch controlling the spool **20** motor, and simultaneously monitor movement of the tarp arms **22** via the truck's mirrors, all while driving, e.g., a 10-ton loaded dump truck **10** over a winding, sloped, dirt

road with a steep drop-off on one side. Further, if the operator shifts attention back to driving but continues to press the rocker switch, the tarp arms **22** will reach the extent of their motion, but the tarp **18** will continue to unwind. When the tarp **18** is fully unwound, if the spool **20** is still turning, it will wind the tarp **18** back on, but in the opposite direction. While the system is fully functional in this “inverted” spool **20** arrangement, the extend/retract action the system will perform is opposite of that on the rocker switch labels. This may confuse the operator, further distracting him or her from the safety-critical task of driving the dump truck **10**.

[0007] The Background section of this document is provided to place embodiments of the present invention in technological and operational context, to assist those of skill in the art in understanding their scope and utility. Unless explicitly identified as such, no statement herein is admitted to be prior art merely by its inclusion in the Background section.

SUMMARY

[0008] The following presents a simplified summary of the disclosure in order to provide a basic understanding to those of skill in the art. This summary is not an extensive overview of the disclosure and is not intended to identify key/critical elements of embodiments of the invention or to delineate the scope of the invention. The sole purpose of this summary is to present some concepts disclosed herein in a simplified form as a prelude to the more detailed description that is presented later.

[0009] According to aspects of the present disclosure, a tarp control system for a dump truck controls a motorized spool to automatically extend and retract the tarp in response to a non-continuous input from a user, in a normal operating mode. A rotary encoder attached to the motorized spool outputs the spool's motion to a controller. Upon receiving a non-continuous (i.e., momentary) user input to, e.g., extend the tarp, the controller controls the motor of the motorized spool to rotate in a direction to release the tarp. As it unwinds from the spool, the tarp is pulled over the open dump truck bed by the rearward-biased tarp arms. The controller tracks the amount of tarp released by monitoring the motorized spool motion. When the tarp reaches a (previously calibrated) fully extended position, the controller stops motion of the motorized spool. An inverse process retracts the tarp automatically in response to a non-continuous user input. The user inputs may comprise a user actuating a soft button displayed on a touchscreen display. The controller may output to the display a graphic presenting a representation of the movement of the tarp between the extended and retracted positions. The tarp control system allows a dump truck operator to momentarily press a single button, then ignore the tarp operation and concentrate on driving or other tasks, as the tarp extends or retracts automatically. The tarp control system eliminates the possibility of the tarp controls becoming reversed due to opposite-direction winding of the tarp on the motorized spool.

[0010] During the calibration procedure, in addition to storing the motorized spool motion for a tarp extension or retraction operation, the duration of operation of the motorized spool is also monitored, and a slightly longer duration is stored. A timer function is reset and started prior to controlling the motorized spool for each automatic tarp extension or retraction operation. If the timer function reaches the stored value, the rotary encoder has failed or the system has lost calibration (e.g., due to tarp stretch). To prevent overheating the motor or other damage, automatic control of the motorized spool is stopped, and the tarp control system enters a failsafe backup mode, where the motorized spool is operated manually in response to continuous user inputs, as in the prior art.

[0011] One aspect relates to a tarp control system for a dump truck. The tarp control system includes a tarp spooled onto a motorized spool attached to the dump truck. The tarp is biased towards an extended position by tarp arms. The tarp is operative to move between extended and retracted positions by control of the motorized spool. The tarp control system includes a rotary encoder configured to be attached to the motorized spool, and a controller. The controller includes non-transitory memory and a user interface. The controller is operatively connected to the rotary

encoder and a motor of the motorized spool. The controller is configured to receive a non-continuous input via the user interface to extend or retract the tarp; control the motor of the motorized spool to turn the spool in one of a first direction to extend the tarp, and a second, opposite direction to retract the tarp; monitor motion of the motorized spool by monitoring an output of the rotary encoder; and automatically stop the spool when the tarp reaches a fully extended or fully retracted position.

[0012] Another aspect relates to a method of automatically controlling a tarp on a dump truck. A non-continuous user input to extend or retract the tarp is received. A motor of a motorized spool holding the tarp is controlled to turn the spool in one of a first direction to extend the tarp, and a second, opposite direction to retract the tarp. Motion of the motorized spool is monitored by monitoring an output of a rotary encoder attached to the motorized spool. The spool is automatically stopped when the tarp reaches a fully extended or fully retracted position.

[0013] Yet another aspect relates to a dump truck. The dump truck includes a bed having an open top, a motorized spool affixed to a first end of the bed, and a tarp wound onto the motorized spool. The dump truck further includes tarp arms affixed to the dump truck and the tarp, and biased toward a second end of the bed opposite the motorized spool. The dump truck also includes a rotary encoder attached to the motorized spool, and a controller having non-transitory memory and a user interface. The controller is operatively connected to the rotary encoder and a motor of the motorized spool, and is configured to receive a non-continuous input via the user interface to extend or retract the tarp; control the motor of the motorized spool to turn the spool in one of a first direction to extend the tarp, and a second, opposite direction to retract the tarp; monitor motion of the motorized spool by monitoring an output of the rotary encoder; and automatically stop the spool when the tarp reaches a fully extended or fully retracted position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which aspects of the disclosure are shown. However, this disclosure should not be construed as limited to the aspects set forth herein. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Like numbers refer to like elements throughout.

[0015] FIG. 1 is a side view of a dump truck with a tarp in a retracted position.

[0016] FIG. 2 is a side view of a dump truck with a tarp between retracted and extended positions.

[0017] FIG. 3 is a side view of a dump truck with a tarp in an extended position.

[0018] FIG. 4 is a block diagram showing a tarp control system in the context of a tarp on a dump truck bed.

[0019] FIG. 5 is a diagram of the internal parts of an optical rotary encoder.

[0020] FIG. 6 is a flow diagram of a method of automatically controlling a tarp on a dump truck.

DETAILED DESCRIPTION

[0021] For simplicity and illustrative purposes, the present disclosure is described by referring mainly to an exemplary aspect thereof. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be readily apparent to one of ordinary skill in the art that the present disclosure may be practiced without limitation to these specific details. In this description, well-known methods and structures have not been described in detail so as not to unnecessarily obscure the present disclosure.

[0022] FIG. 4 shows an example of a tarp control system 30, according to aspects of the present disclosure. The tarp control system 30 comprises a controller 32 and a rotary encoder 34. The rotary encoder 34 is configured to be attached to the shaft of a motorized spool 20 having a tarp 18

wound thereon. The controller **32** receives an output of the rotary encoder **34** to monitor rotary motion of the motorized spool **20**.

[0023] The controller **32** also controls a motor **24** of the motorized spool **20**. As with conventional tarp **18** control, with the tarp arms **22** biasing the leading edge of the tarp towards the rear of the dump truck bed **14**, rotary motion of the motorized spool **20** controls the extension and retraction of the tarp **18** over the dump truck bed **14**. The controller **32** controls tarp **18** extension/retraction by commanding the motor **24** which direction to turn, and for how long—the controller essentially replaces the prior art rocker switch, and automates the process such that the tarp **18** is always extended to the proper length.

[0024] The controller **32** determines how long to command the motor **24** to turn the spool **20** in two complimentary ways. First, in a normal operating mode, the controller **32** monitors the motion of the spool **20** via the rotary encoder **34**, and compares the cumulative motion to a stored value determined during a prior calibration procedure. In one aspect, the rotary encoder **34** is an incremental optical rotary encoder. Second, a timer function **39** within the controller **32** monitors the duration that the motor **24** is controlled to wind or unwind the spool **20**, and terminates the spool **20** motion if the timer function **39** reaches a predetermined target value, which is slightly longer than a normal extend or retract operation should take. The timer function **39** guards against failure of the rotary encoder **34**, or loss of calibration, such as by tarp stretch. If the timer function **39** is triggered, it sends the controller **24** into a failsafe backup mode, wherein automatic control of the motor **24** is terminated, and the controller **24** defaults to manual operation, as in the prior art. These dual control modes provide for robust operation, eliminating the impact of a failure that could otherwise leave the dump truck unable to extend the tarp **18** (and thus violate road regulations) or retract the tarp **18** (which would prevent loading the bed **14**).

[0025] FIG. 5 shows the structure and operation of a simple optical rotary encoder **34**. A light source, such as an LED **50** generates a light beam and directs it towards an encoding disk **52**. The encoding disk **52** is opaque, and includes a plurality of radially spaced translucent or transparent windows **54**. The encoding disk **52** is attached to the shaft of the motorized spool **20**, and turns with the spool **20**. An optical sensor **56**, such as a phototransistor, is arranged on the opposite side of the encoding disk **52** as the LED **50**, and aligned such that light beam from the LED **50** is incident on the sensor **56** when a window **54** allows the light to pass. Between windows **54**, the opaque disk **52** blocks the light. As the motorized spool **20**, and hence the encoding disk **52**, rotates, light from the LED **50** is alternately passed to the sensor **56** and blocked from it, generating a repetitive waveform at the sensor **56** output. Depending on factors such as the number and spacing of windows **54**, speed of rotation of the encoding disk **52**, response time of the sensor **56**, and the like, the repetitive waveform output by the sensor **56** may vary from a sine wave to an almost-square wave. A squaring circuit **58**, such as a Schmitt trigger, converts the output of the sensor **56** to a well-defined square wave, suitable for processing by digital circuitry. In practice, the LED **50** and optical sensor **56** may operate in the infrared portion of the spectrum, such that the rotary encoder **34** is generally unaffected by ambient light in the visible spectrum. Rotary encoders are available that operate other than optically, such as using magnetics, etc. In one aspect, a suitable rotary encoder **34** is the Model HP-38 available from GS Global Resources in Mukwonago, WI.

[0026] The output of the rotary encoder **34** is thus typically a pulse train, wherein each pulse represents a known amount of angular motion, or rotation, of the motorized spool **20**. The frequency of the pulse train indicates the rotary speed of the spool **20**. A quadrature optical rotary encoder includes two optical circuits, and generates two non-aligned waveforms. In this case, the direction of rotation of the spool **20** can be determined by comparison of which pulse train leads the other. However, since the controller **32** directly controls the motor **24**, detection of its direction of rotation by the rotary encoder **34** output is not necessary.

[0027] The tarp control system **30** can be calibrated in several ways. In one aspect, a calibration procedure is performed per pair of rotary encoder **34** and motorized spool **20**, to convert a number

of pulses to a known linear measurement, such as feet, meters, or the like. This conversion factor is stored by the controller **32**. When this tarp control system **30** is installed on a dump truck **10**, the system **30** is calibrated to that truck **10** by measuring the length of the dump truck bed **14** to be covered, and entering that length into the controller **32**. The controller **32** uses the stored conversion factor to convert an ongoing count of pulses during an extension/retraction process into linear measurement, which is compared to the stored bed **14** length. This aspect may be particularly useful where a business or organization operates a fleet of dump trucks, all of which use the same combination of rotary encoder **34** and motorized spool **20**, but have different truck bed **14** lengths. As discussed below, the controller **32** additionally monitors and stores the duration of motor **24** operation when extending and retracting the tarp **18**.

[0028] In another aspect, the tarp control system **30** is calibrated directly in its “native” metric: pulse counts. A user initiates a calibration procedure, such as by selecting an option on the touchscreen display **42** of the controller **32**, by actuating a calibration switch, or by other input. Beginning from the retracted position, the user provides a continuous “extend” input to the controller **32**, which turns the motorized spool **20** to extend the tarp **18** until the user input stops—just like the rocker switch of the prior art. During this process, the controller **32** counts the total number of pulses received from the rotary encoder **34**, and stores that count in memory. Later, during normal operating mode, upon receipt of a momentary input from a user to initiate the process, the controller **32** drives the motor **24** in the proper direction to retract or extend the tarp **18**, while counting the pulses received from the rotary encoder **34**. When the total count reaches the calibrated count stored in memory, the controller **32** stops driving the motor **24**. This aspect may be particularly useful for calibrating individual dump trucks **10**, or when dump trucks **10** in a fleet are equipped with different motorized spools **20** (which may yield different pulse counts per foot, meter, etc.). In this aspect also, the controller **32** additionally monitors and stores the duration of motor **24** operation when extending or retracting the tarp **18**.

[0029] FIG. **4** shows a block diagram of the controller **32**. The controller **32** may for example be implemented with a tablet computer, may be integrated into the dump truck dashboard, may be part of the truck chassis computer system, or the like. The controller **32** includes processing circuitry **38**, timer function **39**, non-transitory memory **40**, Input/Output (I/O) functionality **36**, and optionally (as indicated by dashed lines) a touchscreen display **42**.

[0030] The processing circuitry **38** may comprise one or more hardwired state machines, programmable logic with associated firmware, one or more general purpose microprocessors or Digital Signal Processors with associated software, or any combination of these. The non-transitory memory **40** may comprise register files in the processing circuitry **38**, solid state memory (e.g., DRAM, SRAM, SSD, Flash, etc.), optical media (e.g., CD, DVD), magnetic media, or any other type of non-transitory digital media known in the art. Although depicted as internal to the controller **32**, the non-transitory memory **40** may include external data storage, such as an SD card, external disc drive, or the like.

[0031] As depicted in FIG. **4**, the timer function **39** may be counter circuitry internal to the processing circuitry **38** or a software routine executed on the processing circuitry **38**. Alternatively, the timer function **39** may comprise an external timer that is controlled by the processing circuitry **38**. In one aspect, during the above-described calibration procedure, the processing circuitry **38** additionally records the duration of continuous motor **24** operation. This value is increased by a small factor, such as 10%, and the result is stored as the timer function **39** target value. Upon a user initiating an automatic extension or retraction of the tarp **18**, the processing circuitry **38** resets and starts the timer function **39**. Under normal usage—that is, in a normal operating mode—the controller **24** will stop turning the motor **24** prior to the timer function **39** reaching the target value. However, in case of a rotary encoder **34** malfunction, miscalibration, or other failure, the motor **24** may continue to turn the spool **20** after the tarp **18** should be fully extended or retracted. If allowed to continue, this could result in overheating or other damage to the motor **24**. In this case, when the

timer function **39** reaches the target value, it alerts the processing circuitry **38** that the motor has run for longer than should have been required. Upon this alert, the controller **32** enters a failsafe backup mode. It stops the motor **24** and issues an audible and/or visible alert, such as to the touchscreen display **42**, that the tarp control system **30** has malfunctioned. Thereafter, in the failsafe backup mode, the controller **32** functions as a prior art system, whereby the user inputs—which are momentary, or non-continuous, inputs in normal operating mode—are now continuous inputs. The controller turns the motor **24** for as long as one or the other input is continuously applied. In failsafe backup mode, the dump truck operator must monitor the progress of the tarp arms **22**, and discontinue input to the controller **24** when they reach the extended or retracted position. As described above, the user inputs may be soft buttons on the touchscreen display **42**, or external switches (such as the prior art rocker switch).

[0032] The I/O functionality **36** is configured to receive at least an output of the rotary encoder **34** (i.e., a pulse train signal). The I/O functionality **36** is further configured to output controls to the motor **24** of the motorized spool **20**. For example, the motor control output may comprise two signals (i.e., forward and reverse), wherein a voltage and/or current on one signal drives the motor **24** in one direction, and vice versa (i.e., replacing the functionality of a rocker switch). In some aspects, the I/O functionality **36** additionally receives non-continuous inputs (i.e., momentary button presses) from an operator during normal operating mode, and continuous inputs during failsafe backup mode.

[0033] U.S. Pat. No. 11,377,805 (System and Method for Control of Operational Aspects of a Snow Plow Blade and/or Spreader), by the present Applicant, describes a controller operative to receive user input and control various equipment on a dump truck, such as a hydraulic system and other accessories. The controller may be implemented in a tablet computer mounted in the dump truck cab. The controller may interface to the dump truck, such as by connecting to an OnBoard Diagnostics (OBD) port, or by comprising a node on a Controller Area Network (CAN) bus. User input may be via soft buttons displayed on a touchscreen display, or by repurposing steering-wheel mounted buttons that are typically used to control audio or cruise control. The disclosure of this patent, as well as U.S. patents applications Ser. No. 17/840,197 (Automated Speed-Triggered Bed Lowering System and Method with Manual Override) and Ser. No. 18/119,930 (Automated Dump Truck Tailgate Control), which describe a similar controller implementing other functionality, are incorporated by reference herein, in their entireties.

[0034] In one aspect, the tablet-based controller described in the cited patent and applications also implements the controller **32** described herein. In this aspect (as indicated by a dashed line), the momentary user inputs to initiate automatic extension or retraction of a tarp **18** during normal operating mode (or continuous inputs during failsafe backup mode) comprise the operator tapping (or holding) soft buttons displayed on the touchscreen display **42**. In another aspect (as indicated by a dashed line), the non-continuous or continuous user inputs comprise the actuation of other switches, which are sensed by, or relayed to, the I/O functionality **36** of the controller **32**. As one example, the switches may be steering wheel mounted switches with alternate functionality assigned to them, as described in detail in the '805 patent cited above.

[0035] In these aspects, the touchscreen display **42** may also output a visual indication of the tarp **18** extension and retraction process. For example, the touchscreen display **42** may display an outline of a bar with endpoints labeled “RET” and “EXT”, and the bar fills or empties with color proportionately to the progress of the tarp **18** extension or retraction motion. As another example, the touchscreen display **42** may display a graphic icon representing a dump truck and show the tarp **18** progress during tarp **18** extension or retraction operations.

[0036] The controller **32** has a user interface, but is not limited to a touchscreen display **42** as described above. In one aspect, the controller **32** user interface is simply two pushbutton switches (or the two positions of a rocker switch), which an operator momentarily presses to initiate an automated tarp **18** extension or retraction process in normal operating mode (or holds continuously

in failsafe backup mode).

[0037] FIG. 6 depicts the steps in a method of automatically controlling a tarp **18** on a dump truck **10**. A non-continuous user input to extend or retract the tarp **18** is received (block **102**). A motor **24** of a motorized spool **20** holding the tarp **18** is controlled to turn the spool **20** in one of a first direction to extend the tarp **18**, and a second, opposite direction to retract the tarp **18** (block **104**). Motion of the motorized spool **20** is monitored by monitoring an output of a rotary encoder **34** attached to the motorized spool **20** (block **106**). When the tarp **18** reaches a fully extended or fully retracted position (block **108**), the spool **20** is automatically stopped (block **110**). The decision whether the tarp **18** has reached a fully extended or fully retracted position (block **108**) may be made by maintaining a count of pulses received from the rotary encoder **34**, and comparing the count to a stored value previously determined during a calibration procedure. The stored value may represent a linear measurement (i.e., feet, meters, etc.) or it may simply be a count of pulses.

[0038] Aspects of the present disclosure present numerous advantages over the prior art. By automating the process of extending or retracting a tarp **18** of a dump truck **10**, an operator need not be distracted by continuously monitoring the position of tarp arms **22** while holding down a button such as a rocker switch. This may dramatically improve safety, particularly if the operator attempts to extend or retract the tarp **18** while driving the dump truck **10**. A backup failsafe mode prevents damage in the event of component failure or malfunction, while still allowing for manual extension and retraction of the tarp **18**.

[0039] In some aspects, the tarp control system **30** is easily retrofitted to existing dump trucks **10** by attaching the rotary encoder **34** to a shaft of an existing motorized spool **20**, and replacing wires controlling the motor **24** with motor control outputs of the controller **32**. The controller **32** may be installed in the dashboard of the dump truck cab **12**; integrated into a control bus of the dump truck **10** (e.g., be a CAN bus node); or may comprise a tablet computer mounted in the dump truck cab **12**. In some embodiments, the tarp control system **30** is integrated with a motorized spool **20**, and sold together with the spool **20**, and/or as part of a tarp system (together with tarp arms **22**, mounting hardware, etc.). In some embodiments, the tarp control system **30** is integrated into the design of the dump truck **10** by the original manufacturer.

[0040] The present disclosure may, of course, be carried out in other ways than those specifically set forth herein without departing from essential characteristics of the disclosure. The present aspects are to be considered in all respects as illustrative and not restrictive, and all changes coming within the meaning and equivalency range of the appended claims are intended to be embraced therein.

Claims

1. A tarp control system for a dump truck including a tarp wound onto a motorized spool attached to the dump truck, and biased towards an extended position by tarp arms, and operative to move between extended and retracted positions by control of the motorized spool, the tarp control system comprising: a rotary encoder configured to be attached to the motorized spool; a controller, having non-transitory memory and a user interface, operatively connected to the rotary encoder and a motor of the motorized spool, the controller configured to receive a non-continuous input via the user interface to extend or retract the tarp; control the motor of the motorized spool to turn the spool in one of a first direction to extend the tarp, and a second, opposite direction to retract the tarp; monitor motion of the motorized spool by monitoring an output of the rotary encoder; and automatically stop the spool when the tarp reaches a fully extended or fully retracted position.
2. The system of claim 1 wherein the non-transitory memory of the controller holds calibration data relating motion of the motorized spool to a distance between fully retracted and extended positions.
3. The system of claim 1 wherein the rotary encoder is an incremental encoder that outputs changes in the rotary position of the motorized spool.

4. The system of claim 3 wherein the rotary encoder outputs a series of pulses, each corresponding to a predetermined amount of rotation of the motorized spool.
5. The system of claim 1 wherein the user interface comprises a touchscreen display, and wherein the non-continuous input comprises a user actuating a soft button on the touchscreen display.
6. The system of claim 5 further comprising displaying on the display a graphic representation of the movement of the tarp between the extended and retracted positions.
7. The system of claim 1 wherein the controller further includes a timer function, and wherein controller is further configured to: reset and start the timer function prior to controlling the motor of the motorized spool prior to each tarp extension or retraction operation; and in response to the timer function reaching a predetermined target value that is longer than the duration the motor operates to fully extend or retract the tarp, stop the motor.
8. The system of claim 7 wherein the controller is further configured to, in response to the timer function reaching the predetermined target value, control the motor only in response to receiving continuous inputs via the user interface to extend or retract the tarp.
9. A method of automatically controlling a tarp on a dump truck, comprising: receiving a non-continuous user input to extend or retract the tarp; controlling a motor of a motorized spool holding the tarp, to turn the spool in one of a first direction to extend the tarp, and a second, opposite direction to retract the tarp; monitoring motion of the motorized spool by monitoring an output of a rotary encoder attached to the motorized spool; and automatically stopping the spool when the tarp reaches a fully extended or fully retracted position.
10. The method of claim 9 further comprising, prior to receiving the non-continuous user input to extend or retract the tarp: performing a calibration relating motorized spool motion to tarp position; and storing calibration data relating motion of the motorized spool to a distance between fully retracted and extended positions.
11. The method of claim 9 wherein the user interface comprises a touchscreen display, and wherein the non-continuous input comprises a user actuating a soft button on the touchscreen display.
12. The method of claim 11 further comprising displaying on the display a graphic representation of the movement of the tarp between the extended and retracted positions.
13. The method of claim 9 further comprising: resetting and starting a timer function prior to controlling the motor of the motorized spool prior to each tarp extension or retraction operation; and in response to the timer function reaching a predetermined target value that is longer than the duration the motor operates to fully extend or retract the tarp, stopping the motor.
14. The method of claim 13 further comprising, in response to the timer function reaching the predetermined target value, controlling the motor only in response to receiving continuous inputs via the user interface to extend or retract the tarp.
15. A dump truck, comprising: a bed having an open top; a motorized spool affixed to a first end of the bed; a tarp wound onto the motorized spool; tarp arms affixed to the dump truck and the tarp, and biased toward a second end of the bed opposite the motorized spool; a rotary encoder attached to the motorized spool; and a controller, having non-transitory memory and a user interface, operatively connected to the rotary encoder and a motor of the motorized spool, the controller configured to receive a non-continuous input via the user interface to extend or retract the tarp; control the motor of the motorized spool to turn the spool in one of a first direction to extend the tarp, and a second, opposite direction to retract the tarp; monitor motion of the motorized spool by monitoring an output of the rotary encoder; and automatically stop the spool when the tarp reaches a fully extended or fully retracted position.
16. The dump truck of claim 15 wherein the non-transitory memory of the controller holds calibration data relating motion of the motorized spool to a distance between fully retracted and extended positions.
17. The dump truck of claim 15 wherein the user interface comprises a touchscreen display, and wherein the non-continuous input comprises a user actuating a soft button on the touchscreen

display.

18. The dump truck of claim 17 further comprising displaying on the display a graphic representation of the movement of the tarp between the extended and retracted positions.

19. The dump truck of claim 15 wherein the controller further includes a timer function, and wherein controller is further configured to: reset and start the timer function prior to controlling the motor of the motorized spool prior to each tarp extension or retraction operation; and in response to the timer function reaching a predetermined target value that is longer than the duration the motor operates to fully extend or retract the tarp, stop the motor.

20. The dump truck of claim 19 wherein the controller is further configured to, in response to the timer function reaching the predetermined target value, control the motor only in response to receiving continuous inputs via the user interface to extend or retract the tarp.
